

Skills Summary

Quotients and Remainders in Whole-Number Division

Use this reference while completing your worksheet or when studying.

Skill 1 — Equal groups and arrays

Model: dividing means making equal groups. The **quotient** counts the full groups; the **remainder** is what will not fill another group. Written as $\text{dividend} = \text{divisor} \times \text{quotient} + \text{remainder}$. The remainder is always *smaller* than the divisor.

Example: 23 counters in groups of 4

Step 1. I want full groups of four, so I ring four counters at a time and keep ringing until fewer than four are left.

Step 2. Five rings hold $5 \times 4 = 20$ counters and $23 - 20 = 3$ are outside, and $3 < 4$, so quotient **5**, remainder **3**.

Try it: 30 counters in groups of 4

check: quotient 7, remainder 2

Skill 2 — Check by multiplying back

Rule: $\text{divisor} \times \text{quotient} + \text{remainder} = \text{dividend}$. Take the remainder off the total first, then divide what is left — that also finds a missing number of groups.

Example: $6 \times \square + 2 = 32$

Step 1. The two left over were never put in a group, so I take them off the total before dividing.

Step 2. $32 - 2 = 30$ went into groups of six, and $30 \div 6 = 5$ groups; check $6 \times 5 + 2 = 32$.

Try it: $7 \times \square + 4 = 46$

check: 6

Skill 3 — Estimate the quotient first

Rule: round the dividend to a nearby friendly number that the divisor goes into evenly, then divide. The estimate tells you roughly how big the real quotient should be, so a wildly wrong answer shows up at once.

Example: Estimate $418 \div 8$

Step 1. I only need about how many, so I round the dividend to a hundred that eight divides evenly and use that instead.

Step 2. 418 rounds to **400**, and $400 \div 8 = 50$, so the real quotient is close to fifty.

Try it: Estimate $287 \div 5$

check: round to about 300

Skill 4 — Read what the remainder means

Three ways a story uses the remainder:

- **Round up** — the leftovers still need a place (vans, tents, boxes).
- **Drop it** — the leftovers cannot make another whole thing (bows, teams).
- **The remainder is the answer** — “how many are left over?”

Example: 27 campers, each tent sleeps 6. How many tents?

Step 1. Every camper needs a bed and a tent cannot be split, so the leftovers will still need a tent of their own.

Step 2. $27 \div 6$ gives quotient **4** and remainder **3**; those three still need a tent, so book **5** tents.

Try it: 34 people, each boat holds 6. How many boats?

check: $5 \times 6 = 30$, so **4** boats